

Define autonomy. Write an essay on the relationship between evolution and one or more of autonomy, intelligence, and learning. (2023)

Answer: Autonomy and Its Relationship with Evolution, Intelligence, and Learning

Autonomy:

It means the ability of an agent to act and decide on its own without human help.

Relationship:

- Evolution helps agents become smarter and more independent.
- Learning makes agents improve from experience.
- Intelligence helps them adapt to new situations.
- Together, evolution → intelligence → learning → autonomy.

How do the components of agent programs work? Write pseudocode agent programs for the goal-based and utility-based agents. (2023)

How components of agent program work

1. **Sensors:** পরিবেশ থেকে তথ্য নেয়
2. **Percept:** সেন্সর যা দেখে বা শোনে
3. **Agent Function:** percept দেখে কী action নিতে হবে তা ঠিক করে
4. **Agent Program:** সেই function-এর বাস্তব কোড
5. **Actuators:** কাজ সম্পন্ন করে (যেমন—move, clean)
6. **Environment:** যেখানে এজেন্ট কাজ করে

Goal-Based Agent (Pseudocode)

php Copy code

```
function GOAL_BASED_AGENT(percept):  
    state ← UPDATE_STATE(state, percept)  
    if goal not reached:  
        goal ← FORMULATE_GOAL(percept)  
    action ← SEARCH_FOR_ACTION(state, goal)  
    return action
```

👉 Works to achieve a specific goal.

(একটি নির্দিষ্ট লক্ষ্য পূরণ করে)

Utility-Based Agent (Pseudocode)

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```
function UTILITY_BASED_AGENT(percept):  
    state ← UPDATE_STATE(state, percept)  
    for each action in ACTIONS(state):  
        utility[action] ← EXPECTED_UTILITY(RESULT(state, action))  
    return action with highest utility
```

👉 Chooses the action with highest utility (best result).

(যে কাজটি সবচেয়ে ভালো ফল দেবে সেটি বেছে নেয়)

What is rationality? Differentiate between human agent and artificial agent in terms of rationality. (2018)

Rationality:

Rationality means **doing the right thing** — choosing an action that gives the **best expected outcome** based on what the agent knows.

A rational agent always acts to **maximize performance** or success.

বাংলা:

Rationality মানে সঠিক কাজ করা — অর্থাৎ agent তার জ্ঞান ও তথ্য অনুযায়ী এমন কাজ বেছে নেয় যা সবচেয়ে ভালো ফলাফল দেয়।

একজন rational agent সবসময় নিজের performance সর্বোচ্চ রাখার চেষ্টা করে

Basis	Artificial Agent	Human Agent
Decision	Uses logic and rules only	Influenced by emotions or bias
Knowledge	Stores and processes lots of data	Limited, may forget
Consistency	Always consistent	Can change with mood or experience
Learning	Learns fast with algorithms	Learns slowly from experience
Goal focus	Always goal-oriented	Can get distracted

What is a problem-solving agent? Write the steps of problem solving. (2023)

Problem-Solving Agent: An agent that **solves problems step by step** to reach a goal.

Steps of Problem Solving

1. **Define Problem** – Understand what the problem is.
2. **Choose Strategy** – Decide how to solve it.
3. **Search Actions** – Check possible steps.
4. **Select Best Action** – Pick the best step.
5. **Execute & Monitor** – Do it and see the results.

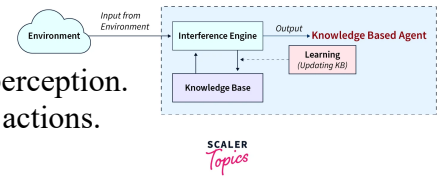
Example: Finding the Shortest Route Home

1. **Define Problem** – Find the fastest way from school to home.
2. **Choose Strategy** – Use a map and compare possible routes.
3. **Search Actions** – Check all roads and transport options.
4. **Select Best Action** – Choose the route with least traffic.
5. **Execute & Monitor** – Take the chosen route and see the time it takes.

Write down a simple algorithm of a generic knowledge-based agent

Algorithm: Knowledge-Based Agent

1. **Start** – Initialize the agent with a **knowledge base**.
2. **Perceive Environment** – Get input from sensors.
3. **Update Knowledge Base** – Add new information from perception.
4. **Infer Actions** – Use logical reasoning to decide possible actions.
5. **Select Action** – Pick the best action according to goals.
6. **Execute Action** – Perform the selected action in the environment.
7. **Repeat** – Go back to Step 2 and continue.



Sure! Here's an easy, exam-ready version in English + Bangla:

Define logical agent. (2021)

English:

A **logical agent** is an agent that **uses knowledge and reasoning** to decide what actions to take. It acts by **deducing actions from facts and rules** in its knowledge base.

Bangla:

Logical agent হলো এমন একটি এজেন্ট যা জ্ঞান ও যুক্তি ব্যবহার করে সিদ্ধান্ত নেয় কোন পদক্ষেপ নেওয়া উচিত। এটি তার **knowledge base**-এ থাকা তথ্য ও নিয়ম থেকে পদক্ষেপ বের করে কাজ করে।

Key points / প্রধান বৈশিষ্ট্য:

1. Uses **logic** to infer actions.
লজিক ব্যবহার করে পদক্ষেপ নির্ধারণ করে।
2. Acts **rationally** based on knowledge.
জ্ঞান অনুযায়ী যুক্তিসঙ্গতভাবে কাজ করে।
3. Can handle **complex environments** if knowledge base is sufficient.
যদি **knowledge base** যথেষ্ট হয়, জটিল পরিবেশও সামলাতে পারে।

বাকীগুলো Note খাতা